AISC 5020 Mathematics & Statistics for AI

Lecture 3 | Linear Algebra II

Gary Choi (CUHK Mathematics)

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(Recall) Data representation

Scalar: A numeric value (e.g., $a = 1$, $b = 2.34$, $c = -5678$)

Vector: an ordered list of numbers of the form $(a_1, a_2, \cdots, a_n)$ or $\begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}$ (e.g., $\begin{pmatrix} 1 & 3 \end{pmatrix}$, $\begin{pmatrix} 3 \\ -1 \\ 4 \\ -1.59 \end{pmatrix}$)

Matrix: A rectangular array (e.g., $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \in \mathbb{R}^{2\times 3}$)

Tensor: a multi-dimensional array for more complex data
Example: A 256×256 color image stored as $\mathcal{X} \in \mathbb{R}^{256\times 256\times 3}$.

In practice, it is common to handle the data as vectors (with suitable reshaping), and so most of our theories and computations are developed based on vectors.

(Recall) Dot product and cosine similarity

For any two d-dimensional vectors $\mathbf{u} = (u_1, u_2, \ldots, u_d), \mathbf{v} = (v_1, v_2, \ldots, v_d) \in \mathbb{R}^d$, we define the **dot product** of them as

$$\boxed{\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + \cdots + u_d v_d = \sum_{j=1}^{d} u_j v_j}$$

We can further define the **cosine similarity** of them using the cosine of the angle between the two vectors:

$$\boxed{\text{cosine similarity} = \cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|_2 \|\mathbf{v}\|_2} = \frac{u_1 v_1 + \cdots + u_n v_n}{\sqrt{u_1^2 + \cdots + u_n^2} \cdot \sqrt{v_1^2 + \cdots + v_n^2}}}$$

(Recall) Norms and distances

The L2 norm is also known as the Euclidean norm, defined as follows: For any n-dimensional vector $\mathbf{x} = (x_1, x_2, \ldots, x_n)$, we have

$$\|\mathbf{x}\|_2 = \sqrt{x_1^2 + \cdots + x_n^2}.$$

The L1 norm is another commonly used norm. For any n-dimensional vector $\mathbf{x} = (x_1, x_2, \ldots, x_n)$, we have

$$\|\mathbf{x}\|_1 = |x_1| + \cdots + |x_n|.$$

The distance between two points $\mathbf{x}, \mathbf{y}$ under any chosen norm can be given by

$$\boxed{d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|}.$$

Topics today

- More on $A\mathbf{x} = \mathbf{b}$: Span, basis, rank, ...
- Eigenvalues and eigenvectors
- Singular value decomposition (SVD)
- Matrix conditioning and sensitivity
- Analyzing graph-structured data

Some key questions about linear systems

Recall that one fundamental task in linear algebra is to solve systems of linear equations:

$$A\mathbf{x} = \mathbf{b}$$

where A is a $m \times n$ matrix, $\mathbf{x}$ is a $n \times 1$ vector, and $\mathbf{b}$ is a $m \times 1$ vector

Some key questions:

- **Existence:** Does there exist such a solution $\mathbf{x}$?
- **Uniqueness:** May there be more than one solution $\mathbf{x}$?
- **Sensitivity:** How much will the solution $\mathbf{x}$ change when the data A, $\mathbf{b}$ change slightly?

Recall the column-combination view

Recall that the matrix-vector multiplication $A\mathbf{x}$ can be viewed as a linear combination of the column vector.

For $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_d]$ and $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{pmatrix}$, we have

$$A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_d\mathbf{a}_d.$$

In other words, the columns are building blocks, and the unknown entries of $\mathbf{x}$ are their coefficients.
Therefore, solving $A\mathbf{x} = \mathbf{b}$ means reconstructing $\mathbf{b}$ from these building blocks via a suitable combination.

Span: The collection of all possible combinations

Mathematically, we define the **span** of a set of vectors as the collection of all possible linear combinations:

$$\text{span}\{\mathbf{a}_1, \ldots, \mathbf{a}_d\} = \left\{ \sum_{j=1}^{d} x_j \mathbf{a}_j : x_j \in \mathbb{R} \right\}.$$

For $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_d]$, we also call the span of its column vectors **the column space of** A, also denoted as $\text{col}(A)$.

- Note that the coefficients $x_1, \ldots, x_d$ can be positive or negative
- The equation $A\mathbf{x} = \mathbf{b}$ is solvable if $\mathbf{b} \in \text{col}(A)$ (i.e., $\mathbf{b} \in \text{span}\{\mathbf{a}_1, \ldots, \mathbf{a}_d\}$)

Example: Let $\mathbf{a}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\mathbf{a}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$. We have $x_1\mathbf{a}_1 + x_2\mathbf{a}_2 = \begin{pmatrix} x_1 + x_2 \\ 2x_1 - x_2 \end{pmatrix}$.

Therefore, $\text{span}\{\mathbf{a}_1, \mathbf{a}_2\} = \left\{ \begin{pmatrix} x_1 + x_2 \\ 2x_1 - x_2 \end{pmatrix} : x_1, x_2 \in \mathbb{R} \right\}$.

In particular, the vector $\mathbf{b} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 2\begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ and so $\mathbf{b} \in \text{span}\{\mathbf{a}_1, \mathbf{a}_2\}$.

Linear independence

Note that the d vectors $\mathbf{a}_1, \ldots, \mathbf{a}_d$ are not necessarily all useful in the representation of the span.

Example: Let $\mathbf{a}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\mathbf{a}_2 = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$. Then $x_1\mathbf{a}_1 + x_2\mathbf{a}_2 = x_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + x_2 \begin{pmatrix} 2 \\ 4 \end{pmatrix} = (x_1 + 2x_2) \begin{pmatrix} 1 \\ 2 \end{pmatrix}$.

So all reachable outputs are still a multiple of $\mathbf{a}_1$, i.e., $\text{span}\{\mathbf{a}_1, \mathbf{a}_2\} = \{c\mathbf{a}_1 : c \in \mathbb{R}\}$.

It represents a straight line on $\mathbb{R}^2$.

To rule out the "hidden redundancy" in the span representation, we recall the concept of **linear independence**.

- A set of vectors $\{\mathbf{a}_1, \mathbf{a}_2, \cdots, \mathbf{a}_d\}$ are **linearly independent** if and only if none of them can be written as a linear combination of the other vectors. Equivalently, $\{\mathbf{a}_1, \mathbf{a}_2, \cdots, \mathbf{a}_d\}$ are linearly independent if and only if

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_d\mathbf{a}_d = 0 \text{ implies } x_1 = x_2 = \cdots = x_d = 0.$$

- If there exists $x_1, \ldots, x_d$ which are **not all zero** such that $\sum_{j=1}^{d} x_j\mathbf{a}_j = 0$, then $\{\mathbf{a}_1, \mathbf{a}_2, \cdots, \mathbf{a}_d\}$ are **linearly dependent**.

Linear independence

Example: For $\mathbf{a}_1 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$, $\mathbf{a}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $\mathbf{a}_3 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$,

- $\mathbf{a}_1$, $\mathbf{a}_2$, $\mathbf{a}_3$ are linearly dependent as

$$\mathbf{a}_3 = \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \mathbf{a}_1 + \mathbf{a}_2.$$

- We can check that $\mathbf{a}_1$, $\mathbf{a}_2$ are linearly independent:

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 = 0 \Rightarrow \begin{pmatrix} 2x_1 + x_2 \\ 3x_1 + x_2 \end{pmatrix} = 0 \Rightarrow \begin{cases} 2x_1 + x_2 = 0 \\ 3x_1 + x_2 = 0 \end{cases} \Rightarrow x_1 = x_2 = 0$$

- Therefore, $\text{span}\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\} = \text{span}\{\mathbf{a}_1, \mathbf{a}_2\} = \left\{ \begin{pmatrix} 2x_1 + x_2 \\ 3x_1 + x_2 \end{pmatrix} : x_1, x_2 \in \mathbb{R} \right\}$.

Basis: linearly independent spanning set

A **basis** is a set of vectors that acts as the minimal building blocks to uniquely construct every vector in a vector space.
More precisely, a set of vectors B is said to be a **basis** for a vector space V if it satisfies both of the following properties:

- **Spanning the Space**: We should have $\text{span}(B) = V$. In other words, we can represent any vector in V using a linear combination of vectors in B.
- **Linear Independence**: B must be linearly independent, i.e., there is no redundancy.

Examples:

- The set $\{(1, 0), (0, 1)\}$ is a basis for $\mathbb{R}^2$ (the standard basis).
- The set $\{(1, 0), (1, 1)\}$ is also a basis for $\mathbb{R}^2$
- The set $\{(1, 0)\}$ is NOT a basis for $\mathbb{R}^2$ as vectors like $(0, 2)$ cannot be represented.
- The set $\{(1, 0), (2, 3), (4, 5)\}$ is NOT a basis for $\mathbb{R}^2$ as the vectors are not linearly independent.

Dimension of a space

Using the concept of basis, we can precisely define the dimension of a space.

Specifically, the **dimension** of a space V is the number of vectors in any basis for it, denoted by

$$\boxed{d = \dim(V)}$$

Examples:

- $\dim(\mathbb{R}^n) = n$
- The dimension of the space of all 2×2 real matrices is $\dim\left(\left\{\begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in R\right\}\right) = 4$, as a basis is $\left\{\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}\right\}$.
- The dimension of the space of all 2×2 real symmetric matrices is $\dim\left(\left\{\begin{pmatrix} a & b \\ b & c \end{pmatrix} : a, b, c \in R\right\}\right) = 3$, as a basis is $\left\{\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}\right\}$.

Rank of a matrix

We define the **rank** of a matrix A as the maximum number of linearly independent row vectors or column vectors it contains. In other words, we have

$$\boxed{\text{rank}(A) = \dim(\text{row space of } A) = \dim(\text{column space of } A) \leq \min(m, n)}$$

for $A \in \mathbb{R}^{m \times n}$.

Here, note that:

- For any matrix A, its column rank is always equal to its row rank. So we only need to check either the maximum number of linearly independent row vectors in A or the maximum number of linearly independent column vectors in A.
- Full column rank means $\text{rank}(A) = n$
- Full row rank means $\text{rank}(A) = m$

Null space

Recall that $\{\mathbf{a}_1, \mathbf{a}_2, \cdots, \mathbf{a}_d\}$ are **linearly dependent** if there exists scalars $x_1, \ldots, x_d$ which are **not all zero** such that

$$\sum_{j=1}^{d} x_j \mathbf{a}_j = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_d\mathbf{a}_d = 0.$$

Therefore, to better study the "redundancy" in all possible linear combinations of $\{\mathbf{a}_1, \mathbf{a}_2, \cdots, \mathbf{a}_d\}$, we consider the **null space** of A, defined by

$$\boxed{\mathcal{N}(A) = \{\mathbf{z} : A\mathbf{z} = 0\}}.$$

The dimension of the null space $\dim(\mathcal{N}(A))$ is called the **nullity** of A.
In particular, note that if $A\mathbf{x} = b$, then

$$A(\mathbf{x} + \mathbf{z}) = b$$

for **every** $\mathbf{z} \in \mathcal{N}(A)$.

Rank-nullity theorem

The **rank-nullity theorem** (also known as the dimension theorem) states that for any matrix $A \in \mathbb{R}^{m \times d}$, the number of columns of A is equal to the sum of the rank of A and the nullity of A. In other words,

$$\boxed{\text{rank}(A) + \dim(\mathcal{N}(A)) = d}.$$

Example: Consider a 3×3 matrix $A = \begin{pmatrix} 1 & 1 & 2 \\ 2 & 0 & 2 \\ 1 & 1 & 2 \end{pmatrix}$

- The number of columns is 3.
- The rank (dimension of the column space) of A is 2, as the third column can be expressed as a linear combination of the first two columns.
- By the rank-nullity theorem, we have

$$\dim(\mathcal{N}(A)) = d - \text{rank}(A) = 3 - 2 = 1$$

A short summary of existence and uniqueness for $A\mathbf{x} = \mathbf{b}$

For $A\mathbf{x} = \mathbf{b}$,

- There exists a solution if $\mathbf{b}$ lies in the column space of A.
- It has a unique solution if the null space only contains 0, i.e., $\mathcal{N}(A) = \{0\}$.
- The rank of A gives the dimension of the column space of A, i.e., the maximum number of linearly independent column vectors it contains.
- The nullity of A gives the dimension of the null space of A (which captures the "redundancy" in all possible linear combinations of the columns of A).
- By the rank-nullity theorem, the number of columns of A is equal to the sum of the rank of A and the nullity of A.

Example

A matrix A has shape 5×3 and rank 3.

- What is the null-space dimension? Can two different $\mathbf{x}$ give the same $A\mathbf{x}$?
- Is every $\mathbf{b} \in \mathbb{R}^5$ reachable?
- What happens to uniqueness if the rank drops to 2?

Example

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- What is the null-space dimension? Can two different $\mathbf{x}$ give the same $A\mathbf{x}$?
- Is every $\mathbf{b} \in \mathbb{R}^5$ reachable?
- What happens to uniqueness if the rank drops to 2?

Answer:

- Nullity is $\#\text{col} - \text{rank}(A) = 3 - 3 = 0$, so $\mathcal{N}(A) = \{0\}$. It follows that

$$A\mathbf{x} = A\mathbf{x}' \Rightarrow A(\mathbf{x} - \mathbf{x}') = 0 \Rightarrow \mathbf{x} - \mathbf{x}' = 0 \Rightarrow \mathbf{x} = \mathbf{x}'$$

  Therefore, two different $\mathbf{x}$ cannot give the same $A\mathbf{x}$.
- The output space has dimension 3 inside $\mathbb{R}^5$; not every $\mathbf{b}$ is reachable.
- If rank drops to 2, nullity becomes $3 - 2 = 1$. Every reachable output has infinitely many coefficient representations.

Matrix decomposition

Many real-world problems involve high-dimensional data!

Examples:

- Healthcare data: Height, weight, blood pressure, heart rate, ...
- Geographical data: Temperature, pressure, rainfall, ...
- Customer data: Income, number of store/website visits, time spent, expenditure, ...
- Stock market data: Trading volume, market capitalization, dividend rate...

This motivates us to consider matrix decomposition: Given a matrix, we want to break it down into some simpler foundational components.

Advantages:

- Uncover hidden patterns / key features of the matrix
- Reduce data size
- Simplify calculations

Eigenvalues and eigenvectors

Given an $n \times n$ matrix A, if we can find a scalar λ and a non-zero vector $\mathbf{v}$ such that

$$\boxed{A\mathbf{v} = \lambda\mathbf{v}}$$

then

- λ is said to be an **eigenvalue** of A
- $\mathbf{v}$ is said to be an **eigenvector** of A associated with λ

Example: For $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$,

$$A\begin{bmatrix} 1 \\ 1 \end{bmatrix} = 3\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad A\begin{bmatrix} 1 \\ -1 \end{bmatrix} = 1\begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

so $\lambda = 3$ is an eigenvalue of A with eigenvector $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$,

and $\lambda = 1$ is another eigenvalue of A with eigenvector $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

Eigenvalues and eigenvectors

Properties: If $A\mathbf{v} = \lambda\mathbf{v}$, then

- $A^k\mathbf{v} = A^{k-1}(A\mathbf{v}) = A^{k-1}(\lambda\mathbf{v}) = \lambda A^{k-1}\mathbf{v} = \cdots = \lambda^k\mathbf{v}$ for all k
- $(A - cI)\mathbf{v} = (\lambda - c)\mathbf{v} \implies \lambda - c$ is an eigenvalue of $A - cI$ (with the same associated eigenvectors $\mathbf{v}$)
- (if A is invertible) $A^{-1}\mathbf{v} = \frac{1}{\lambda}\mathbf{v} \implies \frac{1}{\lambda}$ is an eigenvalue of A^{-1} (with the same associated eigenvectors $\mathbf{v}$)

Finding eigenvalues and eigenvectors

Finding all eigenvalues $\iff$ finding all roots of the **characteristic polynomial**

$$p(\lambda) = \det(A - \lambda I) = 0$$

- $p(\lambda)$ is a degree n polynomial $\rightarrow$ n eigenvalues $\lambda_1, \ldots, \lambda_n$
- Eigenvalues can be real/complex
- It is possible to have repeated eigenvalues

Finding eigenvectors: For each eigenvalue λ_i, solve

$$(A - \lambda_i I)\mathbf{v}_i = 0$$

to get the associated eigenvectors $\mathbf{v}_i$

- For each λ_i, there are infinitely many associated eigenvectors (Note: If $\mathbf{v}_i$ is a solution, then $\alpha\mathbf{v}_i$ is also a solution for any α)
- We usually only need to find one eigenvector for each λ_i

Finding eigenvalues and eigenvectors

Example: For $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, consider

$$\det(A - \lambda I) = \det\left(\begin{bmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{bmatrix}\right) = 0.$$

We have

$$(2-\lambda)^2 - 1 = (\lambda - 3)(\lambda - 1) = 0.$$

Hence, the eigenvalues are $\lambda_1 = 3$ and $\lambda_2 = 1$.

For $\lambda_1 = 3$, $(A - 3I)\mathbf{v} = 0$ gives $v_1 = v_2$, and hence an associated eigenvector is $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

For $\lambda_2 = 1$, $(A - I)\mathbf{v} = 0$ gives $v_1 = -v_2$, and hence an associated eigenvector is $\mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

Diagonalization/eigen-decomposition

- If there exists n linearly independent eigenvectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$ associated with $\lambda_1, \ldots, \lambda_n$, then A is **diagonalizable**:

$$A = VDV^{-1} = \left(\begin{array}{c|c|c|c} \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \end{array}\right) \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} \left(\begin{array}{c|c|c|c} \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \end{array}\right)^{-1}$$

- If $A = VDV^{-1}$, then $A^k = (VDV^{-1})^k = \cdots = VD^kV^{-1}$

Example: For $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, from the previous example, we have already obtained $\lambda_1 = 3$ with an associated eigenvector $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$, and $\lambda_2 = 1$ with an associated eigenvector $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$. Therefore, we have

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}^{-1}$$

Diagonalization/eigen-decomposition

- Not all square matrices are diagonalizable!

  Example: For $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, we have

  $$\det(A - \lambda I) = \lambda^2 = 0 \Rightarrow \lambda_1 = \lambda_2 = 0.$$

  But $(A - 0I)\mathbf{v} = 0$ only has 1 linearly independent solution

  $$\mathbf{v} = \begin{pmatrix} c \\ 0 \end{pmatrix},$$

  where $c \in \mathbb{R}$.

  Therefore, A is not diagonalizable.

Diagonalization/eigen-decomposition

Some theoretical results from linear algebra:

- – If all eigenvalues of A are distinct, then A is diagonalizable.
  (Reason: Eigenvectors corresponding to distinct eigenvalues must be linearly independent.)
  – (**Spectral theorem**) If A is real symmetric, then A is diagonalizable and all eigenvalues of A are real. Moreover, it admits an **orthogonal eigen-decomposition**

$$\boxed{A = QDQ^{\mathsf{T}}},$$

where Q is an orthogonal matrix whose columns are the orthonormal eigenvectors of A.

Example: For $A = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$, we can solve for the eigenvalues and eigenvectors and get:

$$A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 5 \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad A \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

From the above, we get

$$Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \qquad D = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}, \qquad A = QDQ^{\mathsf{T}}.$$

Some useful properties of eigenvalues and eigenvectors

- $\det(A) = \lambda_1 \lambda_2 \cdots \lambda_n$
- If A is upper/lower triangular, then all eigenvalues are the diagonal elements of A
- A is invertible if and only if 0 is not an eigenvalue of A
- If $\mathbf{w} = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n$, then

$$A\mathbf{w} = c_1\lambda_1\mathbf{v}_1 + \cdots + c_n\lambda_n\mathbf{v}_n$$

and

$$A^k\mathbf{w} = c_1\lambda_1^k\mathbf{v}_1 + \cdots + c_n\lambda_n^k\mathbf{v}_n$$

- Consequence:
If there is a dominant eigenvalue λ_i (i.e. $|\lambda_i| > |\lambda_j|$ for all $j \neq i$), then $A^k\mathbf{w} \approx c_i\lambda_i^k\mathbf{v}_i$ as $k \to \infty$

Power method for computing eigenvalues/eigenvectors

Goal: Find an eigenvector associated with the largest (in modulus) eigenvalue of A
Idea: Let $\lambda_1, \lambda_2, \ldots, \lambda_n$ be eigenvalues of A with

$$|\lambda_1| > |\lambda_2| \geq |\lambda_3| \geq \cdots \geq |\lambda_n|$$

(i.e., λ_1 is the dominant eigenvalue of A), and let $\mathbf{v}_1, \ldots, \mathbf{v}_n$ be corresponding eigenvectors. If we pick any vector $\mathbf{b}_0 = c_1\mathbf{v}_1 + c_1\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n$ with $c_1 \neq 0$ (i.e., $\mathbf{b}_0$ has a non-zero component in the direction of $\mathbf{v}_1$), then

$$\begin{aligned} A^k\mathbf{b}_0 &= c_1A^k\mathbf{v}_1 + c_1A^k\mathbf{v}_2 + \cdots + c_nA^k\mathbf{v}_n \\ &= c_1\lambda_1^k\mathbf{v}_1 + c_1\lambda_2^k\mathbf{v}_2 + \cdots + c_n\lambda_n^k\mathbf{v}_n \\ &= c_1\lambda_1^k\left(\mathbf{v}_1 + \frac{c_2}{c_1}\left(\frac{\lambda_2}{\lambda_1}\right)^k\mathbf{v}_2 + \cdots + \frac{c_n}{c_1}\left(\frac{\lambda_n}{\lambda_1}\right)^k\mathbf{v}_n\right) \to c_1\lambda_1^k\mathbf{v}_1 \text{ as } k \to \infty, \end{aligned}$$

since $\frac{|\lambda_j|}{|\lambda_1|} < 1$ for all $j > 1$.

Power method for computing eigenvalues/eigenvectors

In other words, as $k \to \infty$, $A^k \mathbf{b}_0$ will be a scalar multiple of $\mathbf{v}_1$ and hence an eigenvector associated with λ_1.

Power method: for a $n \times n$ matrix A with eigenvalues

$$|\lambda_1| > |\lambda_2| \geq |\lambda_3| \geq \cdots \geq |\lambda_n|,$$

we can compute an approximate eigenvector $\mathbf{w}_1$ of λ_1 by:

1: Set $\mathbf{b}_0$ as an arbitrary vector and N as the prescribed number of iterations.
(Note that the requirement is that $\mathbf{b}_0$ contains a nonzero component in the direction of eigenvector $\mathbf{v}_1$ of λ_1. In practice, we can use a random vector for the initialization.)
2: **for** $k = 1 : N$ **do**
3: $\quad \mathbf{y}_k = A\mathbf{b}_{k-1}$
4: $\quad \mathbf{b}_k = \mathbf{y}_k / \|\mathbf{y}_k\|_2$ (normalization, just to avoid overflow)
5: **end for**
6: $\mathbf{w}_1 = \mathbf{b}_N$ will be the approximate eigenvector of λ_1.

Power method for computing eigenvalues/eigenvectors

Question 1: After obtaining an approximate eigenvector $\mathbf{w}_1$ associated with λ_1, how to get the value of λ_1?

Answer: The value of λ_1 can be obtained by the Rayleigh quotient:

$$\lambda_1 \approx \frac{\mathbf{w}_1^T A \mathbf{w}_1}{\mathbf{w}_1^T \mathbf{w}_1}$$

(Reason: $\frac{\mathbf{w}_1^T A \mathbf{w}_1}{\mathbf{w}_1^T \mathbf{w}_1} \approx \frac{\mathbf{w}_1^T \lambda_1 \mathbf{w}_1}{\mathbf{w}_1^T \mathbf{w}_1} = \lambda_1 \frac{\mathbf{w}_1^T \mathbf{w}_1}{\mathbf{w}_1^T \mathbf{w}_1} = \lambda_1$)

Question 2: What if we want to get the second largest eigenvalue (together with an associated eigenvector) and so on?

Answer: We can consider $\tilde{A} = A - \lambda_1 \mathbf{w}_1 \mathbf{w}_1^T$ (the method is known as *deflation*) and apply the power method to $\tilde{A}$, and repeat the process.

Remark on diagonalization/eigen-decomposition

Eigen-decomposition is good BUT...

- It does not always work!
  - One may consider **Jordan canonical form** $A = VJV^{-1}$, where J is a *block diagonal* matrix
  - However, Jordan canonical form is not very useful in practice as it is highly sensitive to perturbations, e.g., for nonzero ϵ we have

$$A = \begin{bmatrix} \epsilon & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & \epsilon \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & \epsilon \end{bmatrix} \begin{bmatrix} 0 & \epsilon \\ 1 & 1 \end{bmatrix}^{-1}$$

  while

$$A = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}^{-1}$$

- More importantly, it is not applicable to non-square matrices!

A more general method: **Singular value decomposition (SVD)**

Singular value decomposition (SVD)

Goal: Decompose the data matrix A in a way that allows us to easily reduce its dimension / extract information from it.

Consider a data matrix $A \in \mathbb{R}^{n \times d}$, where

- n = number of data points
- d = dimension (number of features) of each data point

Remark: In practice we usually have $n > d$, but the matrix decomposition method would work for general n, d.

Idea:

- We want to achieve a decomposition of the form $\boxed{A = USV^T}$ that works for any (possibly non-square) matrix A
- U, S, V should be analogous to the matrices in the eigen-decomposition
- Two versions: reduced SVD and full SVD

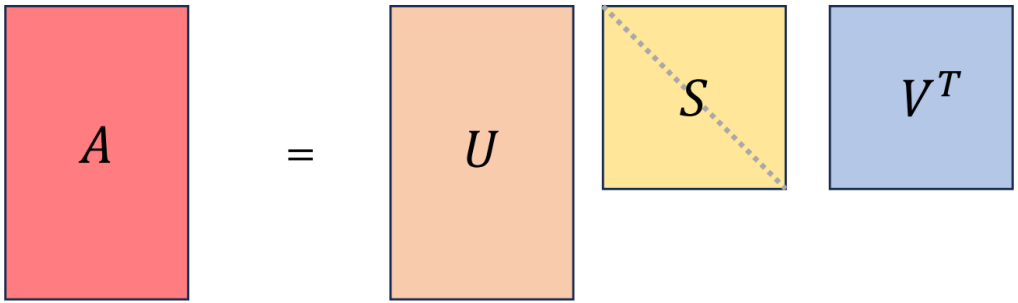
Reduced SVD

Given any $n \times d$ real matrix A, let $k = \min\{n, d\}$. The **reduced SVD** of A is given by

$$\boxed{A = USV^T}$$

where

- U is a $n \times k$ matrix with orthonormal columns ($U^T U = I$)
- S is a $k \times k$ square matrix with diagonal elements $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_k \geq 0$ (called the singular values)
- V is a $d \times k$ matrix with orthonormal columns ($V^T V = I$)



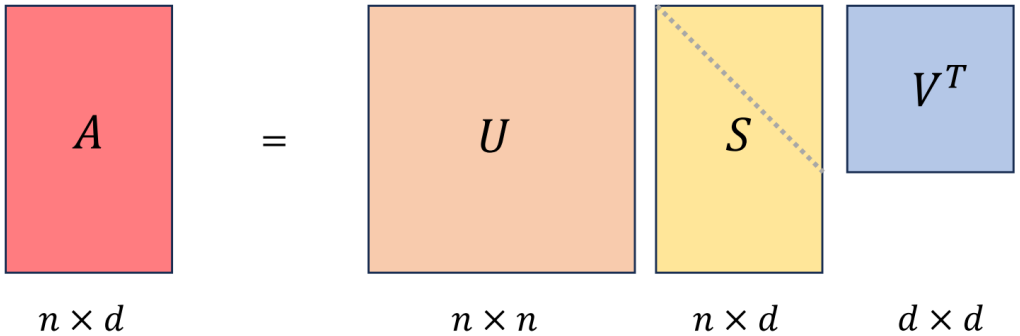
Full SVD

Given any $n \times d$ data matrix A, let $k = \min\{n, d\}$. The **full SVD** of A is given by

$$A = USV^T$$

where

- U is a $n \times n$ orthogonal matrix (i.e., square matrix with orthonormal columns, $U^T U = UU^T = I$)
- S is a $n \times d$ matrix with diagonal (i.e. $s_{11}, s_{22}, \dots$) elements $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_k \geq 0$ (called the singular values)
- V is a $d \times d$ orthogonal matrix (i.e., matrix with orthonormal columns, $V^T V = VV^T = I$)



Full SVD vs Reduced SVD

Full SVD:

- U, V are square matrices (with some columns padded with redundant orthonormal vectors)
- S is a rectangular diagonal matrix (padded with some zeros)

Reduced SVD:

- One of U, V is rectangular (no redundant columns)
- S is a square diagonal matrix

Example: (Full SVD)

$$\underbrace{\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \\ 7 & 8 \end{bmatrix}}_{4\times 2} = \underbrace{\begin{bmatrix} -0.15 & -0.82 & -0.39 & -0.38 \\ -0.35 & -0.42 & 0.24 & 0.80 \\ -0.55 & -0.02 & 0.70 & -0.46 \\ -0.74 & 0.38 & -0.55 & 0.04 \end{bmatrix}}_{4\times 4} \underbrace{\begin{bmatrix} 14.27 & 0 \\ 0 & 0.63 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}}_{4\times 2} \underbrace{\begin{bmatrix} -0.64 & 0.77 \\ -0.77 & -0.64 \end{bmatrix}}_{2\times 2}$$

Example: (Reduced SVD)

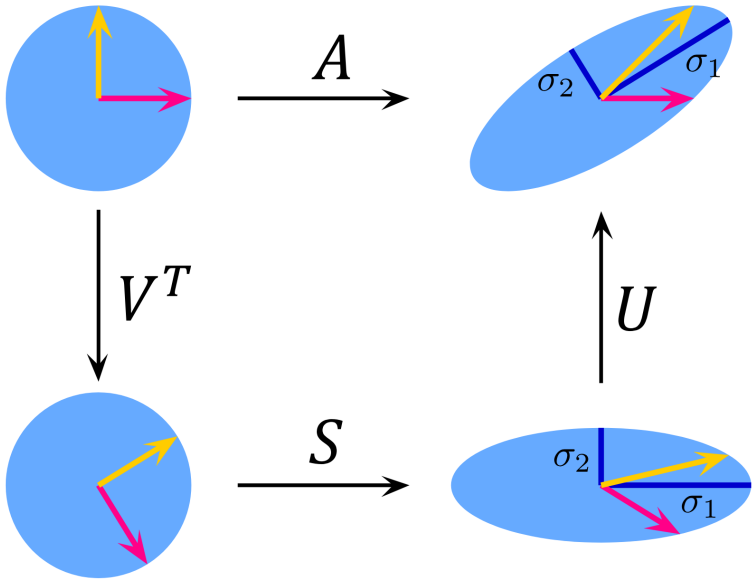
$$\underbrace{\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \\ 7 & 8 \end{bmatrix}}_{4\times 2} = \underbrace{\begin{bmatrix} -0.15 & -0.82 \\ -0.35 & -0.42 \\ -0.55 & -0.02 \\ -0.74 & 0.38 \end{bmatrix}}_{4\times 2} \underbrace{\begin{bmatrix} 14.27 & 0 \\ 0 & 0.63 \end{bmatrix}}_{2\times 2} \underbrace{\begin{bmatrix} -0.64 & 0.77 \\ -0.77 & -0.64 \end{bmatrix}}_{2\times 2}$$

Geometric interpretation of the SVD

For any vector $\mathbf{x} \in \mathbb{R}^d$, we can consider the action of

$$A\mathbf{x} = USV^T\mathbf{x}$$

- V^T: Rotation/Reflection
  (as $\|V^T\mathbf{x}\|_2 = \|\mathbf{x}\|_2$ for all $\mathbf{x}$)
  - Aligns the principal input directions $\mathbf{v}_1, \ldots, \mathbf{v}_d$ align with the standard coordinate axes
- S: Scaling
  - Stretches or shrinks the space along the newly aligned axes by the singular values
- U: Rotation/Reflection
  (as $\|U\mathbf{x}\|_2 = \|\mathbf{x}\|_2$ for all $\mathbf{x}$)
  - Aligns the standard coordinate axes with the principal output directions $\mathbf{u}_1, \ldots, \mathbf{u}_n$



(Image: Wikipedia)

Relationship between SVD and eigen-decomposition

By the spectral theorem in linear algebra, every real symmetric square matrix is diagonalizable, with real eigenvalues and orthonormal eigenvectors.
Now, note that for any real $n \times d$ matrix A,

1. It is easy to check that $A^T A$ is a $d \times d$ real symmetric matrix, and hence we have

$$A^T A = \widetilde{V} \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_d \end{bmatrix} \widetilde{V}^{-1}$$

   for some $d \times d$ orthogonal matrix $\widetilde{V}$ and eigenvalues $\lambda_1, \ldots, \lambda_d$.
   Also, if $(A^T A)\mathbf{v}_i = \lambda_i \mathbf{v}_i$, then

$$\mathbf{v}_i^T A^T A \mathbf{v}_i = \mathbf{v}_i^T (\lambda_i \mathbf{v}_i) = \lambda_i \mathbf{v}_i^T \mathbf{v}_i$$

   and hence $\|A\mathbf{v}_i\|^2 = \lambda_i \|\mathbf{v}_i\|^2 = \lambda_i$. This shows that $\lambda_i \geq 0$ for all i, i.e., all eigenvalues of $A^T A$ are non-negative.

Relationship between SVD and eigen-decomposition

2. Similarly, AA^T is a $n \times n$ real symmetric matrix and hence diagonalizable:

$$AA^T = \widetilde{U} \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix} \widetilde{U}^{-1}$$

for some $n \times n$ orthogonal matrix $\widetilde{U}$. Note that $\lambda_1, \ldots, \lambda_n$ are the same eigenvalues as the corresponding ones in A^TA (except for possibly some zero eigenvalues), as we can see that if $(A^TA)\mathbf{v} = \lambda\mathbf{v}$ for eigenvalue $\lambda \neq 0$ and eigenvector $\mathbf{v}$, then

$$A((A^TA)\mathbf{v}) = A(\lambda\mathbf{v}) \Rightarrow (AA^T)(A\mathbf{v}) = \lambda(A\mathbf{v}),$$

and we have $A\mathbf{v} \neq \mathbf{0}$ (otherwise $\lambda\mathbf{v} = (A^TA)\mathbf{v} = \mathbf{0}$, contradiction).

Also, as in the previous case, all eigenvalues of AA^T must be non-negative.

Relationship between SVD and eigen-decomposition

Now, consider the full SVD of A:

$$\underbrace{A}_{n\times d} = \underbrace{U}_{n\times n}\,\underbrace{S}_{n\times d}\,\underbrace{V^T}_{d\times d}$$

We have

$$\begin{aligned}\underbrace{A^TA}_{d\times d} &= (USV^T)^T(USV^T)\\ &= VS^T\underbrace{U^TU}_{=I}SV^T = VS^TSV^T\\ &= \underbrace{V}_{d\times d}\underbrace{\begin{bmatrix}\sigma_1^2 & & & & & \\ & \ddots & & & & \\ & & \sigma^2_{\min\{n,d\}} & & & \\ & & & 0 & & \\ & & & & \ddots & \\ & & & & & 0\end{bmatrix}}_{d\times d}\underbrace{V^T}_{d\times d}\end{aligned}$$

Relationship between SVD and eigen-decomposition

Therefore, we have

$$\widetilde{V}\begin{bmatrix}\lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_d\end{bmatrix}\widetilde{V}^{-1} = A^TA$$

$$= V\begin{bmatrix}\sigma_1^2 & & & & & \\ & \ddots & & & & \\ & & \sigma^2_{\min\{n,d\}} & & & \\ & & & 0 & & \\ & & & & \ddots & \\ & & & & & 0\end{bmatrix}V^T$$

Since $V^TV = I$, we have $V^T = V^{-1}$, and from the above we have

- $V =$ a set of orthonormal eigenvectors of A^TA
- $\sigma_1 = \sqrt{\lambda_1},\ \sigma_2 = \sqrt{\lambda_2}, \ldots$ (assuming that $\lambda_1, \lambda_2, \ldots$ have been sorted in descending order).

Relationship between SVD and eigen-decomposition

Similarly, by considering

$$
\begin{aligned}
AA^T &= (USV^T)(USV^T)^T \\
&= U(SS^T)U^T \\
&= \underbrace{U}_{n\times n}\underbrace{\begin{bmatrix} \sigma_1^2 & & & & & \\ & \ddots & & & & \\ & & \sigma^2_{\min\{n,d\}} & & & \\ & & & 0 & & \\ & & & & \ddots & \\ & & & & & 0 \end{bmatrix}}_{n\times n}\underbrace{U^T}_{n\times n}
\end{aligned}
$$

and comparing it with the eigen-decomposition of AA^T, we have

- $U =$ a set of orthonormal eigenvectors of AA^T
- $\sigma_1 = \sqrt{\lambda_1},\ \sigma_2 = \sqrt{\lambda_2}, \ldots$ (assuming that $\lambda_1, \lambda_2, \ldots$ have been sorted in descending order).

Relationship between SVD and eigen-decomposition

Conclusion: The matrices U, S, V in the full SVD $A = USV^T$ can be obtained by computing the eigen-decomposition of A^TA and AA^T!

- S: $\sigma_1 = \sqrt{\lambda_1},\ \sigma_2 = \sqrt{\lambda_2}, \ldots$, where all λ_i's are eigenvalues of AA^T or A^TA (sorted in descending order)
- U = a set of orthonormal eigenvectors of AA^T
- V = a set of orthonormal eigenvectors of A^TA

Remark: There are some compatibility conditions for the left singular vectors $\mathbf{u}_i$ in U and the right singular vectors $\mathbf{v}_i$ in V that need to be satisfied. In practice,

- We can first obtain S and V from A^TA as mentioned above.
- Then, we construct U explicitly using

$$\mathbf{u}_i = \frac{A\mathbf{v}_i}{\sigma_i}$$

for all $i = 1, \ldots, \min\{n, d\}$. For the remaining $\mathbf{u}_{\min\{n,d\}+1}, \ldots, \mathbf{u}_n$, we can directly look for some vectors orthonormal to $\mathbf{u}_1, \ldots, \mathbf{u}_{\min\{n,d\}}$.

Low-rank matrix approximation using SVD

By the relationship between SVD and eigen-decomposition, we can see that:

$$\begin{aligned}\text{rank}(A) &= \text{maximum number of linearly independent columns of } A \\ &= \text{maximum number of linearly independent rows of } A \\ &= \text{number of non-zero singular values in the SVD of } A\end{aligned}$$

Therefore, we can use the **truncated SVD** to reduce the dimension of the data matrix A!

- Only keep the first r singular values and set all other singular values as 0
- In other words, we only keep the first r columns of U, the first $r \times r$ submatrix of S and the first r columns of V (i.e. the first r rows of V^T):
  - $\widetilde{U} = U(1:m, 1:r)$
  - $\widetilde{S} = S(1:r, 1:r)$
  - $\widetilde{V} = V(1:n, 1:r)$

Low-rank matrix approximation using SVD

Then, we have

$$\widetilde{A} = \widetilde{U}\widetilde{S}\widetilde{V}^T = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^T + \cdots + \sigma_r \mathbf{u}_r \mathbf{v}_r^T,$$

which gives a rank-r approximation of the original matrix A.

Naturally applicable to data/image/video compression:

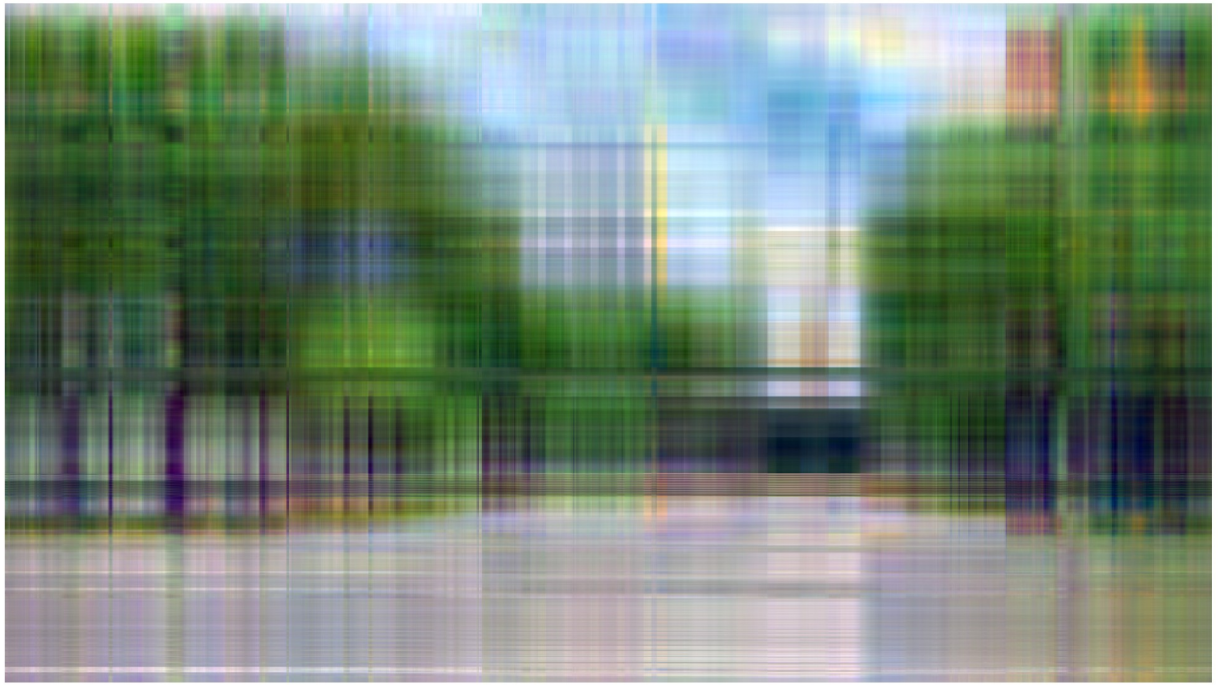
- Keep the most important information
- Require less storage and computation

Example: Image compression



Image: `https://60.cuhk.edu.hk/download/`

Example: Rank-5 approximation



Example: Rank-20 approximation



Example: Rank-100 approximation



Truncated SVD gives an optimal rank-r approximation

Recall that we can get a a rank-r approximation of A using the truncated SVD:

$$\boxed{\widetilde{A} = \widetilde{U}\widetilde{S}\widetilde{V}^T = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^T + \cdots + \sigma_r \mathbf{u}_r \mathbf{v}_r^T}$$

where

- $\widetilde{U} = U(1:m, 1:r)$
- $\widetilde{S} = S(1:r, 1:r)$
- $\widetilde{V} = V(1:n, 1:r)$

Theorem (**Eckart–Young–Mirsky theorem**):
The truncated SVD $\widetilde{A}$ gives an optimal rank-r approximation in the sense that it minimizes the matrix 2-norm (spectral norm) of $A - B$, i.e.,

$$\|A - B\|_2$$

among all matrices $B \in \mathbb{R}^{m \times n}$ with $\text{rank}(B) \leq r$.

(Note: $\|M\|_2 = \max\limits_{\mathbf{x} \neq \mathbf{0}} \dfrac{\|M\mathbf{x}\|_2}{\|\mathbf{x}\|_2} = \max\limits_{\|\mathbf{x}\|_2 = 1} \|M\mathbf{x}\|_2 = \sqrt{\lambda_{\max}(M^T M)} = \sigma_{\max}(M)$.)

Truncated SVD gives an optimal rank-r approximation

Proof:
First, it is easy to see that

$$\|A - \widetilde{A}\|_2 = \left\| \sum_{i=1}^{n} \sigma_i \mathbf{u}_i \mathbf{v}_i^T - \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^T \right\|_2 = \left\| \sum_{i=r+1}^{n} \sigma_i \mathbf{u}_i \mathbf{v}_i^T \right\|_2 = \sigma_{r+1}.$$

Next, we want to show that $\|A - B\|_2 \geq \sigma_{r+1}$ for all $B \in \mathbb{R}^{m \times n}$ with $\text{rank}(B) \leq r$.

For any $B \in \mathbb{R}^{m \times n}$ with $\text{rank}(B) \leq r$, we can write

$$B = XY^T$$

where $X \in \mathbb{R}^{m \times r}$ and $Y \in \mathbb{R}^{n \times r}$ (X can be a matrix whose columns span the column space of B, and Y can be a coefficient matrix).

Truncated SVD gives an optimal rank-r approximation

Proof: (Continued)
Note that the first $r+1$ columns of V (i.e., $\mathbf{v}_1, \ldots, \mathbf{v}_{r+1}$) are orthonormal and span an $(r+1)$-dimensional subspace W, so the linear map $Y^T|_W : W \to \mathbb{R}^r$ gives a linear map from an $(r+1)$-dimensional space into an r-dimensional space $W = \text{span}\{v_1, \ldots, v_{r+1}\} \subset \mathbb{R}^n$.
Any such map has a nontrivial kernel: there exists $\gamma_1, \ldots, \gamma_{r+1}$ not all zero such that

$$w = \gamma_1 \mathbf{v}_1 + \cdots + \gamma_{r+1}\mathbf{v}_{r+1} \in W \setminus \{0\}$$

with

$$Y^T \mathbf{w} = \mathbf{0}.$$

Without loss of generality, we can scale $\mathbf{w}$ so that

$$\|\mathbf{w}\|_2^2 = \gamma_1^2 + \cdots + \gamma_{r+1}^2 = 1.$$

Truncated SVD gives an optimal rank-r approximation

Proof: (Continued)

Then, we have

$$\begin{aligned}
\|A - B\|_2^2 &\geq \|(A - B)\mathbf{w}\|_2^2 \\
&= \|A\mathbf{w} - XY^\top \mathbf{w}\|_2^2 \\
&= \|A\mathbf{w} - \mathbf{0}\|_2^2 \\
&= \|A\mathbf{w}\|_2^2 \\
&= \|\gamma_1 A\mathbf{v}_1 + \cdots + \gamma_{r+1} A\mathbf{v}_{r+1}\|_2^2 \\
&= \|\gamma_1 \sigma_1 \mathbf{u}_1 + \cdots + \gamma_{r+1}\sigma_{r+1}\mathbf{u}_{r+1}\|_2^2 \quad (\text{as } A = USV^T \Rightarrow AV = US \Rightarrow A\mathbf{v}_i = \sigma_i \mathbf{u}_i) \\
&= \gamma_1^2\sigma_1^2 + \cdots + \gamma_{r+1}^2\sigma_{r+1}^2 \quad (\text{as } \mathbf{u}_1, \ldots, \mathbf{u}_{r+1} \text{ are orthonormal}) \\
&\geq (\gamma_1^2 + \cdots + \gamma_{r+1}^2)\sigma_{r+1}^2 \\
&= \sigma_{r+1}^2.
\end{aligned}$$

This completes the proof. ■

SVD and pseudoinverse

While we have introduced SVD primarily as a tool for dimensionality reduction, it is also very useful for solving matrix equations!
Recall that an $n \times n$ matrix A is said to be **invertible** if there exists an $n \times n$ matrix B such that

$$AB = BA = I.$$

The matrix B is unique and is called the **inverse** of A, denoted by $B = A^{-1}$.

Example: For any 2×2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the matrix inverse is

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

In general, the square linear system $A\mathbf{x} = \mathbf{b}$ has a unique solution if and only if A is invertible, which is also equivalent to that $\det(A) \neq 0$

SVD and pseudoinverse

In some cases, $A\mathbf{x} = \mathbf{b}$ does not have a unique solution, and the concept of matrix inverse A^{-1} is **not applicable**. This motivates us to consider developing a generalization of the matrix inverse, called the pseudoinverse.

First, we define:

- Pseudoinverse of scalar: $\sigma^+ = \begin{cases} \frac{1}{\sigma} & \text{if } \sigma \neq 0 \\ 0 & \text{if } \sigma = 0 \end{cases}$
- Pseudoinverse of (possibly rectangular) diagonal matrices:
  $S^+ =$ transpose of the matrix and taking the pseudoinverse of each entry

Then, for any $m \times n$ real matrix A with full SVD $A = USV^T$, we can define the **pseudoinverse** of A as a $n \times m$ matrix

$$\boxed{A^+ = VS^+U^T}$$

SVD and pseudoinverse

The pseudoinverse A^+ shares many similar properties with the standard matrix inverse A^{-1}.

Properties of pseudoinverse:

- If $m = n$ and A is invertible, then $A^+ = A^{-1}$
- If $m \neq n$ and all columns of A are linearly independent, then $A^+A = I$
- If $m \neq n$ and all rows of A are linearly independent, then $AA^+ = I$
- More generally, A^+ always satisfies:

$$AA^+A = A, \quad A^+AA^+ = A^+,$$

$$(AA^+)^T = AA^+, \quad (A^+A)^T = A^+A$$

Solving underdetermined systems using pseudoinverse

One important use of the pseudoinverse is to solve underdetermined systems.

Proposition:
Given any underdetermined system

$$A\mathbf{x} = \mathbf{b}$$

where A is an $m \times n$ matrix with $m < n$ (i.e. more unknowns than equations),
a minimum (2-)norm solution of $A\mathbf{x} = \mathbf{b}$ is

$$\boxed{\mathbf{w} = A^{+}\mathbf{b}},$$

which satisfies $\|\mathbf{w}\|_2 \leq \|\mathbf{x}\|_2$ for all solutions $\mathbf{x}$.

Solving underdetermined systems using pseudoinverse

Proof:
Note that for any $n \times n$ real symmetric matrix M with $M^2 = M$, we can write any $\mathbf{x}$ as $\mathbf{x} = M\mathbf{x} + (I - M)\mathbf{x}$ and get

$$\begin{aligned}
\|\mathbf{x}\|^2 =& \mathbf{x}^\mathsf{T}\mathbf{x} \\
=& \Big(M\mathbf{x} + (I - M)\mathbf{x}\Big)^\mathsf{T}\Big(M\mathbf{x} + (I - M)\mathbf{x}\Big) \\
=& \mathbf{x}^\mathsf{T}M^\mathsf{T}M\mathbf{x} + \mathbf{x}^\mathsf{T}M^\mathsf{T}(I - M)\mathbf{x} + \mathbf{x}^\mathsf{T}(I - M)^\mathsf{T}M\mathbf{x} + \mathbf{x}^\mathsf{T}(I - M)^\mathsf{T}(I - M)\mathbf{x} \\
=& \|M\mathbf{x}\|^2 + \|(I - M)\mathbf{x}\|^2 + \mathbf{x}^\mathsf{T}\Big(M^\mathsf{T} + M - 2M^\mathsf{T}M\Big)\mathbf{x} \\
=& \|M\mathbf{x}\|^2 + \|(I - M)\mathbf{x}\|^2,
\end{aligned}$$

where the last equality holds since $M^\mathsf{T} = M$ and $M^2 = M$.

Solving underdetermined systems using pseudoinverse

Proof: (Continued)
Since A^+A is symmetric and

$$(A^+A)^2 = (A^+A)(A^+A) = (A^+AA^+)A = A^+A,$$

we can take $M = A^+A$. Then, for any solution $\mathbf{x}$ of $A\mathbf{x} = \mathbf{b}$, from the result established above, we have

$$\begin{aligned}\|\mathbf{x}\|^2 &= \|A^+A\mathbf{x}\|^2 + \|(I - A^+A)\mathbf{x}\|^2 \\ &= \|A^+(A\mathbf{x})\|^2 + \|(I - A^+A)\mathbf{x}\|^2 \\ &= \|A^+\mathbf{b}\|^2 + \|(I - A^+A)\mathbf{x}\|^2 \\ &\geq \|A^+\mathbf{b}\|^2.\end{aligned}$$

It is also easy to check that $A(A^+\mathbf{b}) = \mathbf{b}$ and hence $A^+\mathbf{b}$ is a solution of $A\mathbf{x} = \mathbf{b}$. We conclude that $A^+\mathbf{b}$ is a minimum-norm solution. ■

Conditioning of matrices

The study of conditioning of matrices arises from the observation that even if the square linear system $A\mathbf{x} = \mathbf{b}$ is solvable, there may still be some practical issues regarding the stability.
Example: Consider $A\mathbf{x} = \mathbf{b}$ with

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 10^{-4} \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 10^{-4} \end{bmatrix}.$$

It is easy to see that the solution is $\mathbf{x} = (1, 1)^{\mathsf{T}}$.
However, if we slightly perturb the second entry in $\mathbf{b}$ by 10^{-4}, we have

$$\mathbf{b}' = \begin{bmatrix} 1 \\ 2 \cdot 10^{-4} \end{bmatrix}, \qquad \mathbf{x}' = (1, 2)^{\mathsf{T}}.$$

A tiny change in the data in this system can produce a significant change in the solution!

To measure how sensitive the solution is to small changes or errors in the input data, we introduce the concept of **condition number**.

Condition number of a matrix

For an invertible square matrix A, the **condition number** of A is given by

$$\boxed{\kappa_2(A) = \|A\|_2 \left\|A^{-1}\right\|_2},$$

where the matrix 2-norm $\|A\|_2$ is defined as

$$\|A\|_2 = \max_{\|x\|_2=1} \|Ax\|_2 .$$

Using the singular values of A, we can further simplify the expression as

$$\boxed{\kappa_2(A) = \frac{\sigma_{\max}}{\sigma_{\min}} \geq 1}$$

Example: For $A = \begin{bmatrix} 2 & 0 \\ 0 & 10^{-4} \end{bmatrix}$, we have $\kappa_2(A) = \dfrac{2}{10^{-4}} = 20000.$

Condition number and relative perturbation bound

Let A be a given invertible matrix. Consider

$$A\mathbf{x} = \mathbf{b} \qquad \text{and} \qquad A(\mathbf{x} + \delta\mathbf{x}) = \mathbf{b} + \delta\mathbf{b}.$$

We have $A(\delta\mathbf{x}) = \delta\mathbf{b}$, which implies that

$$\|\delta\mathbf{x}\|_2 \leq \left\|A^{-1}\right\|_2 \|\delta\mathbf{b}\|_2 .$$

Also, $\|\mathbf{b}\|_2 \leq \|A\|_2 \|\mathbf{x}\|_2$. Therefore, for $\mathbf{b} \neq 0$, we have $\boxed{\dfrac{\|\delta\mathbf{x}\|_2}{\|\mathbf{x}\|_2} \leq \kappa_2(A)\dfrac{\|\delta\mathbf{b}\|_2}{\|\mathbf{b}\|_2}}$

This shows how the condition number of A can be used for estimating the relative perturbation of the solution $\mathbf{x}$ when the input $\mathbf{b}$ is perturbed.

Remarks:

- The above bound is only an inequality. Errors aligned with strongly observed directions may be mild; those aligned with weak directions can be greatly amplified.
- Altogether, a large condition number signals potential sensitivity. However, it **does NOT** prove that every solution computed from the matrix is inaccurate by that factor.

Example

For $A = \begin{bmatrix} 5 & 0 \\ 0 & 0.01 \end{bmatrix}$:

- Find $\kappa_2(A)$.
- If $\delta\mathbf{b} = (0, 0.002)^\mathsf{T}$, find $\delta\mathbf{x}$.
- What is $\kappa_2(A^\mathsf{T}A)$?
- Does using an SVD solver eliminate the measurement error?

Example

For $A = \begin{bmatrix} 5 & 0 \\ 0 & 0.01 \end{bmatrix}$:

- Find $\kappa_2(A)$.
- If $\delta\mathbf{b} = (0, 0.002)^{\mathsf{T}}$, find $\delta\mathbf{x}$.
- What is $\kappa_2(A^{\mathsf{T}}A)$?
- Does using an SVD solver eliminate the measurement error?

Answer: We have

$$\kappa_2(A) = 500,$$
$$\delta\mathbf{x} = A^{-1}\delta\mathbf{b} = (0, 0.2)^{\mathsf{T}},$$
$$\kappa_2(A^{\mathsf{T}}A) = 250000.$$

From the above, using an SVD solver does not eliminate measurement error.

A summary on solving $A\mathbf{x} = \mathbf{b}$

To solve

$$A\mathbf{x} = \mathbf{b}$$

where A is a $m \times n$ matrix, $\mathbf{x}$ is a $n \times 1$ vector, and $\mathbf{b}$ is a $m \times 1$ vector, we can have the following strategies:

- If it is a square system, use a suitable factorization-based method (e.g., the LU factorization).
- If it is a overdetermined system: No actual solution; We can solve for the least-squares solution using the QR factorization method.
- If it is a underdetermined system: Infinitely many solutions; Use SVD and pseudoinverse to solve for a minimum-norm solution.
- To study the sensitivity of the solution to small changes or errors in the input data, check the condition number of A.

Analyzing graph-structured data using linear algebra

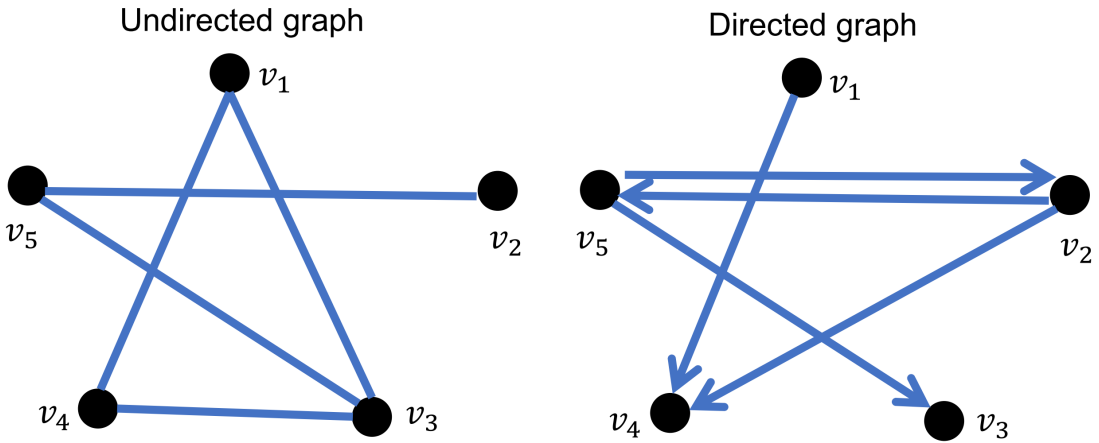
Why are we interested in **graph-structured data**?

- Focus on the connectivity between data items
  - Biological networks
  - Social network users
  - ...
- Can model the movement of information
- Reveal the importance of certain vertices

Graph theory basics

A **graph/network** $G = (V, E)$ consists of:

- A set of vertices V (representing the samples, sensors or variables)
- A set of undirected/directed edges E (representing relationships or similarity weights between the vertices)



Graph theory basics

Some other terminologies:

- A loop (also known as self-loop) is an edge that connects a vertex to itself.
- multiple edges (also known as parallel edges) are edges that have the same end nodes.
- A simple graph is a graph that has no loops and no multiple edges.

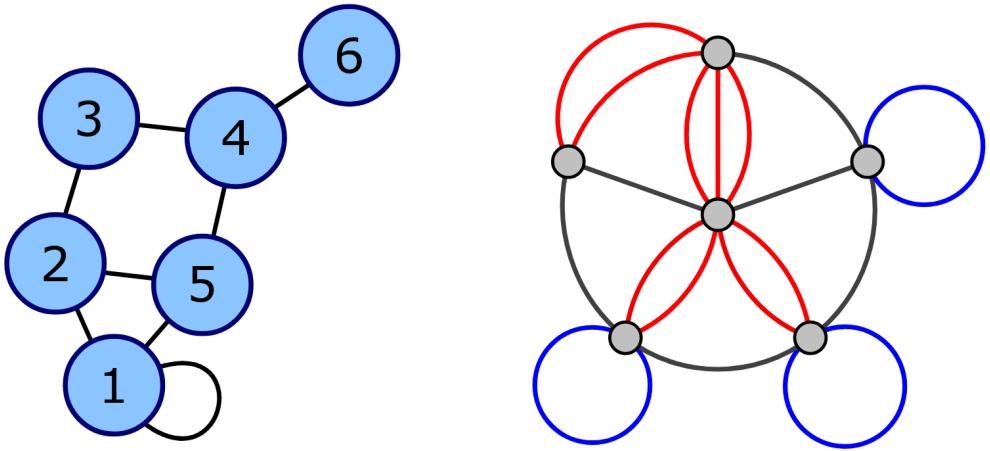


Image source: Wikipedia

Adjacency matrix

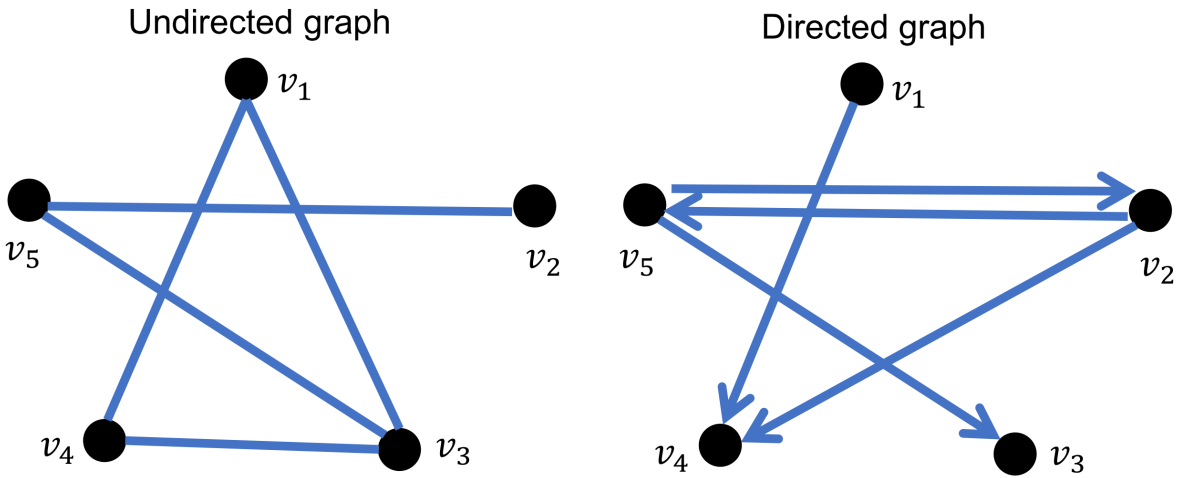
The **adjacency matrix** A of a simple graph G is a $|V| \times |V|$ matrix with

$$A_{i,j} = \begin{cases} 1 & \text{if there is a (undirected/directed) edge from } v_i \text{ to } v_j \\ 0 & \text{otherwise} \end{cases}$$

Note:

- For undirected graph, A must be symmetric
- For directed graph, A may not be symmetric
- If the graph is not a simple graph, we can define the adjacency matrix using the total number of edges between each pair of vertices
- More generally, we can also consider weighted graphs in which the edges can have different weights. In this case, we can have $A_{i,j} = w_{i,j}$ where $w_{i,j}$ is the prescribed weight on the edge connecting v_i and v_j

Example



$$\begin{bmatrix} 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{bmatrix}$$

Incidence matrix

A vertex is incident to an edge if the vertex is one of the two vertices that the edge connects.

For undirected graph, the incidence matrix is a $|V| \times |E|$ matrix B with

$$B_{i,j} = \begin{cases} 1 & \text{if vertex } v_i \text{ is incident to edge } e_j \\ 0 & \text{otherwise} \end{cases}$$

For directed graph, we have

$$B_{i,j} = \begin{cases} 1 & \text{if edge } e_j \text{ enters vertex } v_i \\ -1 & \text{if edge } e_j \text{ leaves vertex } v_i \\ 0 & \text{otherwise} \end{cases}$$

Vertex degree and degree matrix

The degree of a vertex v (denoted as $\deg(v)$) is the number of edges that are incident to v.

The degree matrix of a graph is an $|V| \times |V|$ matrix D with

$$\begin{cases} D_{i,i} = \deg(v_i) & \text{for all } i, \\ D_{i,j} = 0 & \text{if } i \neq j. \end{cases}$$

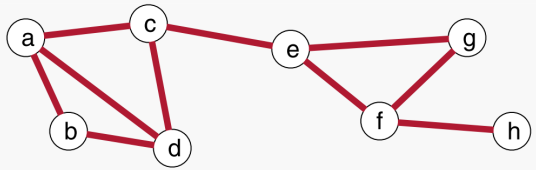
Some properties:

- **Degree sum formula** (also known as the handshaking lemma): For any undirected graph $G = (V, E)$, we have
$$\sum_{v \in V} \deg(v) = 2|E|.$$
- For undirected graph, the degree of a vertex is the corresponding row sum of the adjacency matrix.
- For directed graph, we can further consider row sum and column sum of the adjacency matrix as the out-degree $\deg^-(v)$ and in-degree $\deg^+(v)$ of a vertex v respectively.

Example

Example: Undirected Graph

Consider graph $G = (V, E)$ with the following vertices: $V = \{a, b, c, d, e, f, g, h\}$ and edges $E = \Big\{\{a, b\}, \{a, c\}, \{a, d\}, \{b, d\}, \{c, d\}, \{c, e\}, \{e, f\}, \{e, g\}, \{f, g\}, \{f, h\}\Big\}$.



Alternatively, G can be represented as a matrix with 1 if there is an edge, and 0 otherwise.

$$\begin{array}{c|cccccccc} & a & b & c & d & e & f & g & h \\ \hline a & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ b & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ c & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ d & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ e & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ f & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ g & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ h & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{array} = \begin{pmatrix} 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix}.$$

Source: Mathematical Foundations for Data Analysis

Example

Example: Adjacency and Degree Matrices

Again using our example graph, we can define the adjacency A and degree matrix D:

$$A = \begin{pmatrix} 0&1&1&1&0&0&0&0 \\ 1&0&0&1&0&0&0&0 \\ 1&0&0&1&1&0&0&0 \\ 1&1&1&0&0&0&0&0 \\ 0&0&1&0&0&1&1&0 \\ 0&0&0&0&1&0&1&1 \\ 0&0&0&0&1&1&0&0 \\ 0&0&0&0&0&1&0&0 \end{pmatrix} \quad D = \begin{pmatrix} 3&0&0&0&0&0&0&0 \\ 0&2&0&0&0&0&0&0 \\ 0&0&3&0&0&0&0&0 \\ 0&0&0&3&0&0&0&0 \\ 0&0&0&0&3&0&0&0 \\ 0&0&0&0&0&3&0&0 \\ 0&0&0&0&0&0&2&0 \\ 0&0&0&0&0&0&0&1 \end{pmatrix}.$$

Source: Mathematical Foundations for Data Analysis

Walks, paths, and cycles

- A **walk** from v_0 and v_m in G is a sequence of vertices

$$P = (v_0, v_1, \ldots, v_m)$$

  such that $\{v_i, v_{i+1}\} \in E$ for all $i = 0, \ldots, m-1$.
- The **length** of P is m (as it contains m edges in total).
- If there are no repeated vertices in P, we call P a **path**.
- If there are no repeated vertices in P and $v_0 = v_m$, we call P a **cycle**.

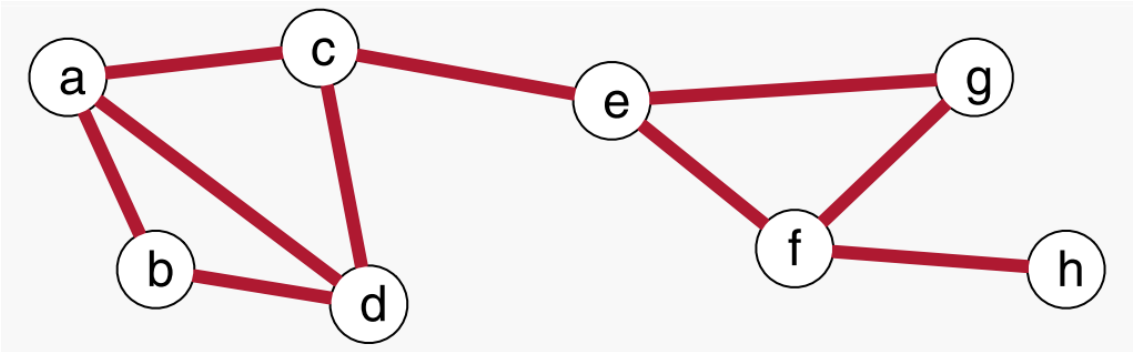
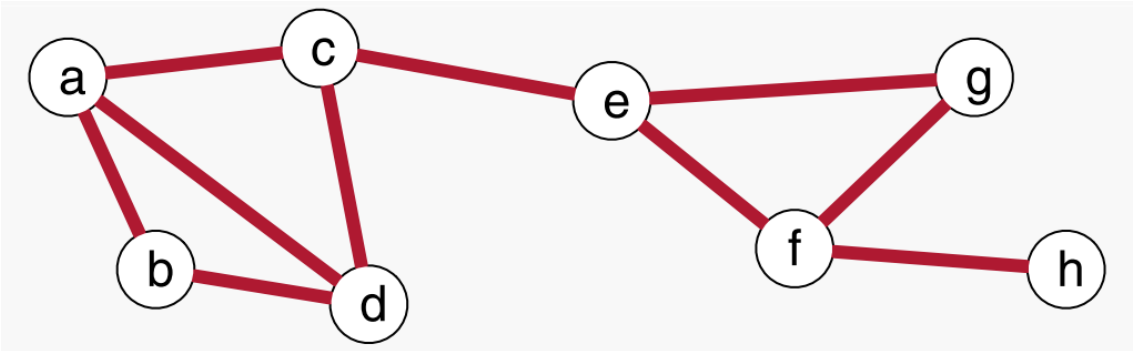


Image source: Mathematical Foundations for Data Analysis

Connectedness, connected components, and graph distances

- We say that v_i and v_j are **connected** if there exists a walk from v_i to v_j.
- We say that the graph G is **connected** if for every $v_i, v_j \in G$, v_i and v_j are connected.
- A **connected component** of G is a maximal connected subgraph of G.
- The **graph distance** $\boldsymbol{d}_E$ (induced by the edge set E) between two vertices v_i and v_j is the minimum number of edges required to get from v_i to v_j.

Example: For the graph below, the distance $\boldsymbol{d}_E(a, g) = 3$, realized by the edges $\{a, c\}$, $\{c, e\}$, and $\{e, g\}$.



Graph Laplacian

We now introduce the concept of graph Laplacian, which plays an important role in the analysis of the structure of a graph.

Let $G = (V, E)$ be an undirected graph. The (unnormalized) **Laplacian** (also known as the Laplace matrix) of G is a $|V| \times |V|$ matrix L with

$$L_{ij} = \begin{cases} \deg(v_i) & \text{if } i = j, \\ -1 & \text{if } v_i \text{ and } v_j \text{ are adjacent,} \\ 0 & \text{otherwise.} \end{cases}$$

Equivalently, we have

$$\boxed{L = D - A},$$

where D is the degree matrix and A is the adjacency matrix of G.

Remark: For directed graphs, we usually either

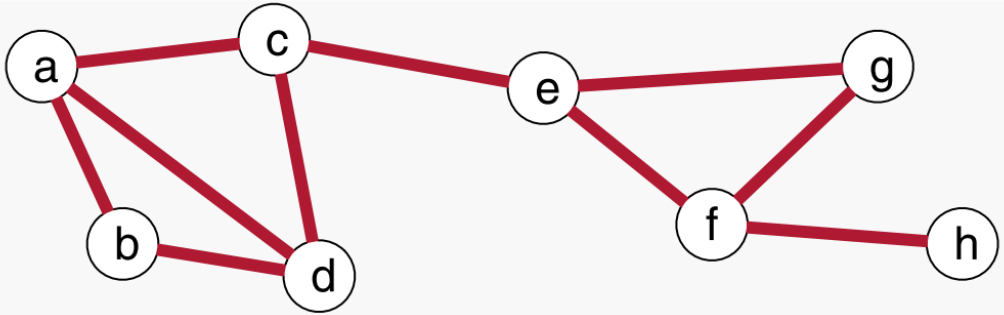
- use in/out-degree to formulate the in/out-degree Laplacian, or
- symmetrize the graph by treating all edges as undirected to proceed with the analysis

Example

Example: Laplacian matrix

The *Laplacian matrix* of our example graph is

$$L_0 = D - A = \begin{pmatrix} 3 & -1 & -1 & -1 & 0 & 0 & 0 & 0 \\ -1 & 2 & 0 & -1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 3 & -1 & -1 & 0 & 0 & 0 \\ -1 & -1 & -1 & 3 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 3 & -1 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 3 & -1 & -1 \\ 0 & 0 & 0 & 0 & -1 & -1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 1 \end{pmatrix}.$$



Source: Mathematical Foundations for Data Analysis

Alternative formulation of the Laplacian

Alternatively, the Laplacian L of an undirected graph G can also be expressed as

$$\boxed{L = \widetilde{B}\widetilde{B}^{\top}},$$

where $\widetilde{B}$ is a $|V| \times |E|$ oriented incidence matrix of G.
More specifically, we simply assign an arbitrary direction for each edge in G and construct the corresponding incidence matrix $\widetilde{B}$ (in the sense of directed graph). For instance, we can set

$$\widetilde{B}_{i,k} = \begin{cases} 1 & \text{if } e_k = (v_j, v_i), \\ -1 & \text{if } e_k = (v_i, v_j), \\ 0 & \text{otherwise.} \end{cases}$$

Spectrum of a graph

Properties of the Laplacian:

- For undirected graphs, since both A and D are real symmetric, L is also real symmetric $\Rightarrow$ diagonalizable
- Moreover, since $L = \widetilde{B}\widetilde{B}^\top$, it is positive semi-definite
  $\Rightarrow$ all eigenvalues are nonnegative real numbers

Hence, we can study the structure of G via eigen-decomposition. This field of study is known as spectral graph theory.

For simplicity, we usually sort all eigenvalues of the Laplacian L in ascending order:

$$\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_{|V|}$$

The eigenvalues are also called the **spectrum** of G.

Spectral properties of graphs: λ_1

Property: For any graph G, the smallest eigenvalue is $\lambda_1 = 0$.

Proof: It is easy to check that the row sum for each row of the Laplacian L is 0:

$$\sum_{j=1}^{|V|} L_{ij} = \deg(v_i) + (-1) \cdot (\#\text{ of vertices adjacent to } v_i) = 0.$$

Therefore, 0 is an eigenvalue of L with the vector of ones $\mathbf{1}$ being an associated eigenvector:

$$L\mathbf{1} = \mathbf{0} = (0)\mathbf{1}$$

Since λ_1 is the smallest eigenvalue, we have $\lambda_1 = 0$. ■

Spectral properties of graphs: λ_2 and beyond

We now move on to the second-smallest eigenvalue λ_2.

Property: For any graph G,

(i) The second-smallest eigenvalue $\lambda_2 > 0$ if and only if G is connected.

(ii) More generally, the number of zero eigenvalues of L is equal to the number of connected components of G.

(iii) The spectrum of G is the union of the spectra of its connected components.

Spectral properties of graphs: λ_2 and beyond

Proof:

(For (i) and (ii))
Let k be the number of connected components in G, and denote the connected components as $G_1, G_2, \ldots, G_k$.

We consider vectors $\boldsymbol{u}^{(1)}$, $\boldsymbol{u}^{(2)}$, $\ldots$, $\boldsymbol{u}^{(k)}$ with

$$\boldsymbol{u}^{(i)}(j) = \begin{cases} 1 & \text{if } j \in G_i, \\ 0 & \text{otherwise.} \end{cases}$$

Then we have

- $\boldsymbol{u}^{(i)} \cdot \boldsymbol{u}^{(j)} = 0$ for all $i \neq j$.
- $L\boldsymbol{u}^{(i)} = \mathbf{0}$ for all i.

This shows that there are at least k mutually orthogonal eigenvectors corresponding to the eigenvalue 0, and hence the number of zero eigenvalues is $\geq k$.

Spectral properties of graphs: λ_2 and beyond

We then further prove that the number of zero eigenvalues is $\leq k$. Note that $(L\boldsymbol{u})(i) = \sum_{j:(i,j)\in E}(\boldsymbol{u}(i) - \boldsymbol{u}(j))$ and hence

$$\boldsymbol{u}^\top L\boldsymbol{u} = \sum_{(i,j)\in E}(\boldsymbol{u}(i) - \boldsymbol{u}(j))^2,$$

from which we can see that $\boldsymbol{u}^\top L\boldsymbol{u} = 0$ if and only if $\boldsymbol{u}(i) = \boldsymbol{u}(j)$ for all $(i,j) \in E$. Therefore, $\boldsymbol{u}$ should have equal value for all vertices in a connected component.
Consequently, if $\boldsymbol{u}$ is an eigenvector associated with $\lambda = 0$, then

$$L\boldsymbol{u} = 0 \Rightarrow \boldsymbol{u}^\top L\boldsymbol{u} = 0 \Rightarrow \boldsymbol{u} = \sum_i \alpha_i \boldsymbol{u}^{(i)}$$

for some $\alpha_1, \ldots, \alpha_k$. Hence every eigenvector of L corresponding to eigenvalue 0 is in span$\{\boldsymbol{u}^{(1)}, \boldsymbol{u}^{(2)}, \ldots, \boldsymbol{u}^{(k)}\}$, which shows that the number of zero eigenvalues is exactly k.

Spectral properties of graphs: λ_2 and beyond

(For (iii))
Note that we can relabel and group the vertices based on the k connected components $G_1, \ldots, G_k$. We can then express the Laplacian L as a block diagonal matrix:

$$L = \begin{bmatrix} L_1 & & & \\ & L_2 & & \\ & & \ddots & \\ & & & L_k \end{bmatrix}$$

where L_i is the Laplacian of G_i. As the eigenvalues of a block diagonal matrix are the union of the eigenvalues of its individual diagonal submatrices, the result follows.

■

Spectral properties of graphs: λ_2 and beyond

From the above result, we see that λ_2 gives some useful information about the connectivity of the graph G. More generally, the value of λ_2 tells us how well connected G is:

- A large λ_2 means that the graph is "very connected".
- A small λ_2 means that although G is connected, there may be certain parts with only a few edges, and missing those edges will already make the graph disconnected.

λ_2 is also known as the **algebraic connectivity** or the **Fiedler eigenvalue** of the graph (named after mathematician Miroslav Fiedler 1926–2015).

Normalized Laplacian

For large graphs, the vertex degrees may be large and pose challenges in our computation. This motivates us to consider the **normalized Laplacian** $\tilde{L}$, which is also an $|V| \times |V|$ matrix but with the following entries:

$$\tilde{L}_{i,j} = \begin{cases} 1 & \text{if} \quad i = j, \\ \dfrac{-1}{\sqrt{\deg(v_i) \cdot \deg(v_j)}} & \text{if} \quad i \neq j \text{ and } (v_i, v_j) \in E, \\ 0 & \text{otherwise.} \end{cases}$$

Equivalently, we have

$$\boxed{\tilde{L} = (D^+)^{1/2} L (D^+)^{1/2} = I - (D^+)^{1/2} A (D^+)^{1/2}},$$

where D^+ is the pseudoinverse of the degree matrix D. In other words, D^+ is also a diagonal matrix but with entries

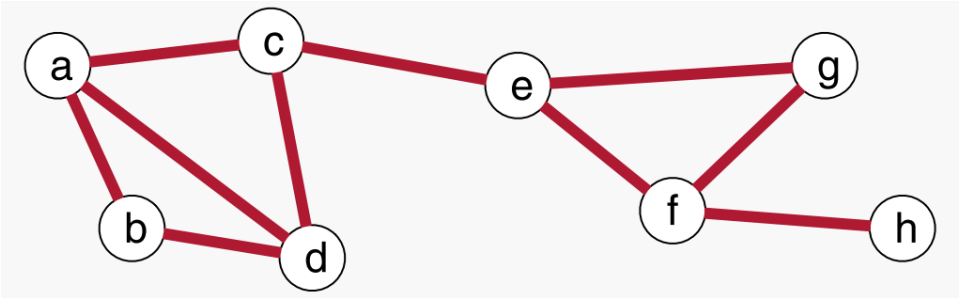
$$(D^+)_{ii} = \begin{cases} 1/D_{ii} & \text{if } D_{ii} \neq 0 \\ 0 & \text{if } D_{ii} = 0. \end{cases}$$

Example

Example: Normalized Laplacian

The *normalized Laplacian* matrix of our example graph is $L = I - D^{-1/2}AD^{-1/2} =$

$$\begin{pmatrix} 1 & -0.408 & -0.333 & -0.333 & 0 & 0 & 0 & 0 \\ -0.408 & 1 & 0 & -0.408 & 0 & 0 & 0 & 0 \\ -0.333 & 0 & 1 & -0.333 & -0.333 & 0 & 0 & 0 \\ -0.333 & -0.408 & -0.333 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -0.333 & 0 & 1 & -0.333 & -0.408 & 0 \\ 0 & 0 & 0 & 0 & -0.333 & 1 & -0.408 & -0.577 \\ 0 & 0 & 0 & 0 & -0.408 & -0.408 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -0.577 & 0 & 1 \end{pmatrix}.$$



Source: Mathematical Foundations for Data Analysis

Some other results and bounds on the spectrum

Property: For $G = (V, E)$ with $n = |V| \geq 2$, let $\lambda_1, \ldots, \lambda_n$ be the spectrum obtained from the normalized Laplacian $\tilde{L}$. We have:

(i) $\lambda_1 + \lambda_2 + \cdots + \lambda_n = n$.

(ii) $\lambda_2 \leq \dfrac{n}{n-1} \leq \lambda_n$.

Proof:

(i) We have sum of all eigenvalues $= \mathrm{tr}(\tilde{L}) = n$.

(ii) From (i) and the fact that $\lambda_1 = 0$, we have

$$n = \lambda_2 + \cdots + \lambda_n \geq (n-1)\lambda_2 \Rightarrow \lambda_2 \leq \frac{n}{n-1}$$

$$n = \lambda_2 + \cdots + \lambda_n \leq (n-1)\lambda_n \Rightarrow \lambda_n \geq \frac{n}{n-1}$$



Some other results and bounds on the spectrum

Remark: If we use the unnormalized Laplacian here (i.e., $\lambda_1, \ldots, \lambda_n$ are the spectrum obtained from the unnormalized Laplacian L), then we have

$$\lambda_1 + \lambda_2 + \cdots + \lambda_n = \sum_{i=1}^{n} \deg(v_i) = 2|E|$$

and

$$\lambda_2 \leq \frac{\sum_{i=1}^{n} \deg(v_i)}{n-1} = \frac{2|E|}{n-1} \leq \lambda_n.$$

More on λ_2

We give some further explanations on why λ_2 can be used for assessing how well connected G is.

Given a graph G, define:

- The volume of a cluster $\text{vol}(S)$ as the number of edges with at least one vertex in S
- The cut between two clusters S, T, $\text{cut}(S, T)$, as the number of edges with one vertex in S and the other in T

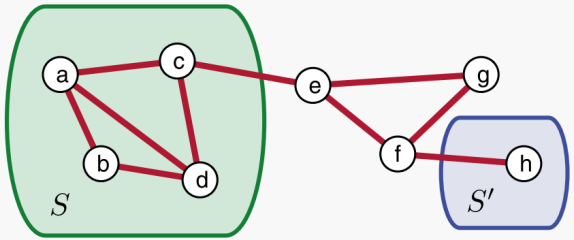
The normalized cut between S and T is defined as

$$\text{ncut}(S, T) = \frac{\text{cut}(S, T)}{\text{vol}(S)} + \frac{\text{cut}(S, T)}{\text{vol}(T)}$$

Example

Example: Normalized Cut

In the running example graph, clusters $S = \{a, b, c, d\}$ and the singleton cluster with $S' = \{h\}$ both have a small cut value with both $\mathsf{cut}(S', \bar{S}') = 1$ and $\mathsf{cut}(S, \bar{S}) = 1$. The volumes however are very different with $\mathsf{vol}(S) = 6$ and $\mathsf{vol}(\bar{S}) = 5$ while $\mathsf{vol}(S') = 1$ and $\mathsf{vol}(\bar{S}') = 10$.



The difference in volumes shows up in the normalized cut score for S and S'. Specifically $\mathsf{ncut}(S', \bar{S}') = 1 + \frac{1}{10} = 1.1$, whereas $\mathsf{ncut}(S, \bar{S}) = \frac{1}{6} + \frac{1}{5} = 0.367$. Overall S results in the smallest normalized cut score; this aligns with how it intuitively is the partition which best divides the vertices in a balanced way without separating along too many edges.

Source: Mathematical Foundations for Data Analysis

More on λ_2: Cheeger's inequality

We further define the **conductance** of a set S in $G = (V, E)$ as

$$\phi(S) = \frac{\text{cut}(S, V \setminus S)}{\text{vol}(S)}$$

Now, for the entire graph G, we define the **conductance** of G as the smallest conductance among all sets with at most half of the total volume:

$$\phi(G) = \min_{S:\text{vol}(S) \leq \text{vol}(V)/2} \phi(S)$$

We have the following important result in spectral graph theory:

Theorem (Cheeger's Inequality)
For any graph G, we have

$$\boxed{\frac{\lambda_2}{2} \leq \phi(G) \leq \sqrt{2\lambda_2}}$$

where λ_2 is the second-smallest eigenvalue of the normalized Laplacian of G

More on λ_2: Cheeger's inequality

Implications of Cheeger's inequality:

- Fundamentally relates a structural property of a graph (its connectivity) to an algebraic property (the eigenvalues of its Laplacian matrix)
- A graph with a small "spectral gap" (in terms of the value of λ_2) must have a sparse cut with few edges relative to the size of the vertex subsets it separates, i.e., a "'bottleneck" (in terms of $\phi(G)$ or the normalized cut)

Next time

Vector Calculus and Computation Graphs

- Derivatives, gradients, Jacobian, Hessians
- Multivariable chain rule
- Taylor series
- Computation graphs
- Automatic differentiation